8(a)(i)	Polarisation is where the **oscillations in a wave are confined to one direction only in a plane** normal to the direction of transfer of energy of the wave.	**B1**
(ii)	**The student is wrong**. Sound waves are **longitudinal** in nature. The **particles** in the medium (usually air) **vibrate parallel to the direction of wave propagation** and hence cannot be polarised.	**A1** **M1**
(iii)	Intensity = $\underline{2I_0}$	**A1**
(iv)	Using Malu's Law, $I_1 = I_0 \cos^2(30° - 15°)$ $I_2 = I_1 \cos^2(50° - 30°)$ $= I_0 \cos^2(15°) \cos^2(20°)$ $= 0.82387 I_0$ $= \underline{0.82 I_0}$ (2 sf)	**C1** **A1**
(b)(i)	distance between slits, d = 1 / (4.5 x 10^5) = 2.222 x 10^{-6} m tan θ = 0.60 / 2.0 θ = 16.699° For diffraction grating, $d \sin \theta = n\lambda$ λ = 2.222 x 10^{-6} sin 16.699° = 640 nm	**C1** **M1**
(ii)	Assume θ = 90° to find maximum number of bright spots. (2.222 x 10^{-6}) sin 90° = n (640 x 10^{-9}) n = 3.47 This means there are 3 bright spots above the central spot, and 3 bright spots below the central spot. In all, there will be 3 + 3 + 1 = $\underline{7 \text{ bright spots}}$.	**C1** **A1** **A1**

(iii)	A **bright white fringe is observed at the centre** of the screen, because **all wavelengths of light reach the screen with zero path difference**. The **other bright fringes consist of a continuous spectrum of colours**, because **different wavelengths of light diffracted at different angles**. For every spectrum of colours, **violet colour is always closer to the centre and red is the furthest away**. This is because **shorter wavelengths diffract at smaller angles** OR **longer wavelengths diffract at larger angles** *1 mark can be awarded if 2 of the observations are correctly made but the explanations are inadequate.*	**B1** **B1** **B1**
(c)(i)	From the graph, the distance between two successive bright fringes is **5.0×10^{-3} m**. For double slit experiment, $a = \lambda D/x$ $= (640 \times 10^{-9})(2.0) / (5 \times 10^{-3})$ $= \underline{2.56 \times 10^{-4}\ \text{m}}$ (2 sf)	**C1** **A1**
(ii)	From the graph, X shows the second constructive interference. This means that the path difference of the two waves at X is 2λ. Hence, path difference $= 2.0 \times 640 \times 10^{-9}$ $= \underline{1.3 \times 10^{-6}\ \text{m}}$ (2 sf)	**A1**
(iii)(A)	Destructive interference takes place at Y. Hence when one of the slits is partially blocked by the tracing paper, the **light wave's intensity / amplitude will be decreased**. As a result, at Y, **there will not be complete destructive interference**. **The dark fringe at Y will now be brighter**.	**B1** **A1**
(iii)(B)	**There will not be any change** (distance between fringes is not affected by amplitude / intensity of light coming through the slits).	**A1**